

Evaluation Framework

Metrics, RAGAS Integration, and Benchmarking Design

RAG-Based Energy Forecasting

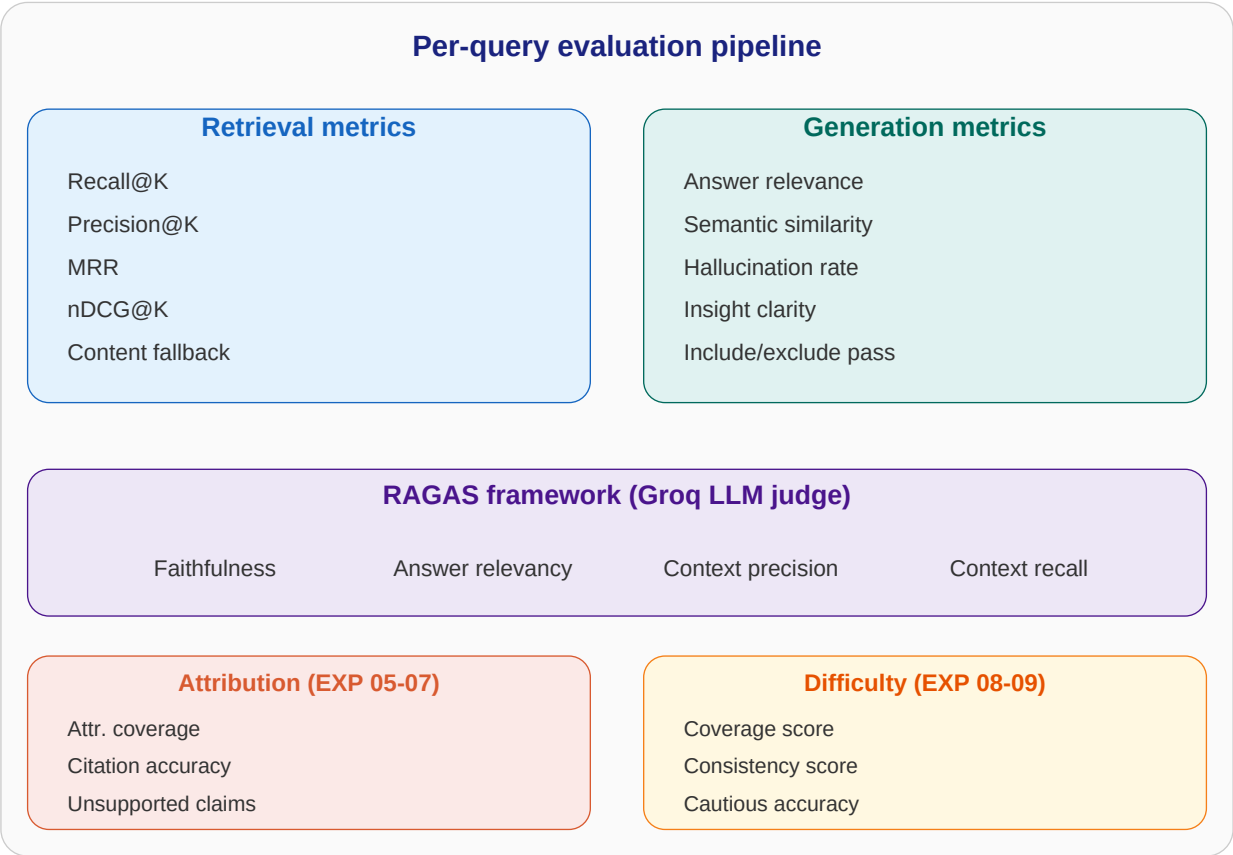
MSc Thesis — LJMU

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1. Overview

This document describes the evaluation framework used across all 10 experiments. The framework combines custom retrieval metrics, custom generation quality metrics, and RAGAS (Retrieval-Augmented Generation Assessment) framework metrics to provide a comprehensive assessment of each pipeline variant.

2. Framework Diagram



3. Custom Retrieval Metrics

Metric	Formula	Interpretation
Recall@K	$\text{relevant_retrieved} / \text{relevant_available}$	What fraction of expected chunks were found?
Precision@K	$\text{relevant_retrieved} / K$	What fraction of retrieved chunks are relevant?
MRR	$1 / \text{rank_of_first_relevant}$	How high is the first relevant chunk ranked?
nDCG@K	$\text{DCG@K} / \text{IDCG@K}$	Rank-weighted retrieval quality (penalises low-ranked hits)

3.1 Content Fallback Scoring

The golden dataset contains approximately 10 queries whose zone labels do not match any KB chunk (Zone 20, 21 references). Pure ID matching always scores these as zero. The content fallback mechanism uses SequenceMatcher text overlap to detect semantically matching chunks even when IDs differ. Threshold: 0.55 overlap ratio.

4. Custom Generation Metrics

Metric	Method	Purpose
Answer Relevance	Cosine similarity (query vs answer embedding)	Does the answer address the question?
Semantic Similarity	Cosine similarity (answer vs reference embedding)	How close is the answer to the ground truth?
Hallucination Rate	Sentence-level entailment check	What fraction of answer claims lack context support?
Insight Clarity	Heuristic (length, structure, specificity)	Is the answer well-structured and informative?
Include/Exclude Pass	Keyword matching	Does the answer contain required terms and avoid forbidden terms?

5. RAGAS Framework Metrics

RAGAS 0.4.x is used with Groq (Llama 3.3 70B) as the LLM judge. Four RAGAS metrics are computed per query in batched evaluation with key rotation.

RAGAS Metric	What It Measures	LLM Calls
Faithfulness	Are answer claims supported by retrieved context?	2-3 per row
Answer Relevancy	Is the answer relevant to the question?	1-2 per row
Context Precision	Is retrieved context relevant to the question?	1 per row
Context Recall	Does retrieved context cover the ground truth?	1 per row

5.1 RAGAS Configuration

- LLM wrapper: LangchainLLMWrapper(ChatGroq) with key rotation
- Embeddings: LangchainEmbeddingsWrapper(all-MiniLM-L6-v2)
- answer_relevancy.strictness = 1 (Groq requires n=1)
- Batch size: 2 rows per batch, 10s delay between batches
- Subsample: 50 rows per experiment (stratified random, seed=42)

6. Attribution Metrics (Experiments 05-07)

Metric	Formula	Purpose
Attribution Coverage	$\text{claims_with_citation} / \text{total_claims}$	What fraction of answer claims cite evidence?
Citation Accuracy	$\text{correct_citations} / \text{total_citations}$	What fraction of citations reference valid evidence?
Spurious Citations	Count of [En] not in evidence set	How many citations reference non-existent evidence?
Unsupported Claim Rate	$\text{uncited_claims} / \text{total_claims}$	What fraction of claims lack any citation?

7. Difficulty-Aware Metrics (Experiments 08-09)

Metric	Formula	Purpose
Coverage Score	$\text{above_threshold} / \text{total_scores}$	Fraction of retrieval scores above confidence threshold
Consistency Score	$1 - \text{std}(\text{scores}) / \text{mean}(\text{scores})$	How consistent are the retrieval scores?
Difficulty Score	$0.6 * \text{coverage} + 0.4 * \text{consistency}$	Combined query difficulty estimate
Cautious Accuracy	$\text{correct_caution} / \text{hard_queries}$	Does the model express uncertainty on hard queries?

8. Benchmarking Design

All experiments are benchmarked under identical conditions to ensure fair comparison:

- Same dataset: combined_golden_dataset_200.csv (200 queries)
- Same LLM: Llama 3.3 70B Versatile via Groq API
- Same temperature: 0 (deterministic)
- Same max_tokens: 500
- Same embedding model: all-MiniLM-L6-v2
- Same evaluation metrics across all experiments
- Only the retrieval pipeline changes between experiments

9. Experiment Matrix

Group	Experiments	Enhancement	Key Metric
Baselines	EXP_01, 02, 03, 04	None / Retrieval variants	Recall@K, Faithfulness
Attribution	EXP_05, 06, 07	Evidence-tagged citation grounding	Attribution Coverage
Difficulty	EXP_08, 09	Query difficulty + adaptive prompting	Cautious Accuracy
Analysis	EXP_10	Comparative benchmarking	Composite Ranking